RECALLING Article 28(b) of the Convention on the International Maritime Organization con cerning the functions of the Committee, NOTING that the International Convention for the Safety of Life at Sea ("the Convention"), 1974 and the International Code for the Constr uction and Equipment of Ships Carrying Liquefied Gases in Bulk ("the IGC Code") currentl y do not specifically provide requirements for carriage of liquefied hydrogen in bulk by sea, NOTING ALSO that paragraph 5 of Preamble of the IGC Code states that requirements f or new products and their conditions of carriage will be circulated as recommendations, on an --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolu tions/MSCResolutions/MSC.566(109).pdf new products and their conditions of carriage will be circulated as recommendations, on an interim basis, prior to the entry into force of the appropriate amendments, RECOGNIZING a need for the development of the Revised interi m recommendations for carriage of liquefied hydrogen in bulk, ACKNOWLEDGING that, in the interim, there is an urgent need to provide recommendations to the Administrations on sa fe carriage of liquefied hydrogen in bulk, ACKNOWLEDGING ALSO that the Revised Interim R ecommendations are intended to facilitate establishment of a tripartite agreement for a pilot ship, which will be developed for the research and demonstration of safe long-dist ance overseas carriage of liquefied hydrogen in bulk, HAVING CONSIDERED the Revised Inte rim Recommendations prepared by the Sub - Committee on Carriage of Cargoes and Container s at its ninth session, 1 ADOPTS the Revised interim recommendations for carriage of liq uefied hydrogen in --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Ind exofIMOResolutions/MSCResolutions/MSC.566(109).pdf 1 ADOPTS the Revised interim recommen dations for carriage of liquefied hydrogen in bulk, the text of which is set out in the annex to the present resolution; 2 INVITES Member States to apply the Revised Interim Re commendations to the pilot ship carrying liquefied hydrogen in bulk taking the explanato ry notes into consideration; 3 AGREES to acquire information on safe carriage of liquefi ed hydrogen in bulk prior to amendment to the IGC Code for the inclusion of liquefied hy drogen; 4 ALSO AGREES that these Revised Interim Recommendations may need to be reviewed if they are to be applied to ships other than the pilot ship; and 5 URGES Member States and the industry to submit information, observations, comments and recommendations based on the practical experience gained through the application of the Revised Interim Recomm endations and submit relevant safety analysis on ships carrying liquefied hydrogen in bu lk.

6 REVOKES resolution MSC.420(97).

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.566(109).pdf on ships carrying liquefied hydrogen in bulk.

6 REVOKES resolution MSC.420(97).

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 2 I:\MSC\108\MSC 108 -20-Add.1.docx ANNEX REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK 1 INTRODUCTION 1.1 For the carriage of liquefied gases in bulk by ships, the shi ps should comply with the relevant requirements in the IGC Code, as amended ("the Code") .

The scope of the Code provided in paragraph 1.1.1 is: "The Code applies to ships regardl ess of their size, including those of less than 500 gross tonnage, engaged in the carria ge of liquefied gases having a vapour pressure exceeding 0.28 MPa absolute at a temperat ure of 37.8°C, and other products, as shown in chapter 19, when carried in bulk".

1.2 A ship carrying liquefied hydrogen in bulk (hereinafter called "liquefied hydrogen - -- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/M SCResolutions/MSC.566(109).pdf products, as shown in chapter 19, when carried in bulk".

1.2 A ship carrying liquefied hydrogen in bulk (hereinafter called "liquefied hydrogen c arrier") should comply with the Code.

1.3 The Code requires that a gas carrier should comply with the minimum requirements for the cargo listed in chapter 19.

However, the requirements for liquefied hydrogen are not specified in the Code.

1.4 This annex provides the Revised Interim Recommendations, as referred to in paragraph 5 of the preamble of the Code, for the carriage of liquefied hydrogen in bulk, which are intended to provide the basis for the future minimum requirements for the carriage of th is cargo.

The Revised Interim Recommendations are intended to facilitate the establishme nt of a t

ripartite agreement among the relevant Administrations for the carriage of liquefied hyd
rogen in bulk.

However, they are not intended to prohibit the adoption of designs and arrangements ---
URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCR
esolutions/MSC.566(109).pdf in bulk.

However, they are not intended to prohibit the adoption of designs and arrangements othe
r than those specified in the Code or in these Recommendations, at the discretion of the
Administrations.

1.5 These recommendations have been developed under the assumption that a liquefied hydr
ogen carrier does not carry liquefied gases other than liquefied hydrogen.

These recommendations, therefore, are not applicable to liquefied hydrogen carriers carr
ying gase s other than liquefied hydrogen.

1.6 In the Code, reference is made to paragraph 5 of the Preamble; paragraph 1.1.6.1; an
d Note No.8 on completion of certificate in "model form of international certificate of
fitness for the carriage of liquefied gases in bulk" in appendix 2 to the Code.

1.7 These Revised Interim Recommendations consist of the following parts.

Part A is applicable to ships with any type of cargo containment system.

Part B and subsequent part(s) --- URL: https://wwwcdn.imo.org/localresources/en/Knowledg
eCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf applicable to ships with a
ny type of cargo containment system.

Part B and subsequent part(s) prescribe additional special requirements for cargo contai
nment systems of specific types.

Part A: General (applicable to ships with any type of cargo containment system); Part B:
Cargo containment systems of independent cargo tanks using vacuum insulation; and Part C
: Cargo containment systems of independent cargo tanks using insulation materials and hy
drogen gas in the inner insulation spaces.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR
RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 3 I:\MSC\108\MSC 108
-20-Add.1.docx 1.8 Part A of this document was developed based on the design of parts B
and C.

If subsequent part(s) are added, the special requirements prescribed in part A may be re
viewed.

Part A General (Applicable to ships with any type of cargo containment system) 2 GENERAL
2.1 Definition --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Indexof
IMOResolutions/MSCResolutions/MSC.566(109).pdf reviewed.

Part A General (Applicable to ships with any type of cargo containment system) 2 GENERAL
2.1 Definition 2.1.1 The following definition should apply for the purpose of these Revi
sed Interim Recommendations.

Permeation is flow of a fluid through another material by diffusion without a defect or
opening of the latter.8 2.2 Requirements for carriage of liquefied hydrogen in bulk 2.2.
1 The requirements for the carriage of liquefied hydrogen in bulk have been developed ba
sed on the results of a comparison study of similar cargoes listed in chapter 19 of the
Code, e.g.

liquefied natural gas.

2.2.2 Chapter 19 of the Code governs the application of general requirements for respect
ive cargoes.

Selections of the general requirements for respective cargoes are expressed in columns '
c' to 'g'.

In addition to general requirements, special requirement s may apply to specific cargoes
depending on the properties/hazards of the cargoes.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.566(109).pdf specific cargoes depending on the properties/hazards of
the cargoes.

2.2.3 Tables 1 and 2 specify the proposed selection of the general requirements and the
special requirements, respectively, for liquefied hydrogen.

In addition to table 2, special requirements for cargo containment systems of specific t
ypes are prescribed in part B or subsequent part(s).

8 1 See paragraph 3.79 of ISO/TR 15916:2015.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 4 I:\MSC\108\MSC 108 -20-Add.1.docx Table 1: Requirements for carriage of liquefied hydrogen in bulk a b c d e f g h i Product name Ship type Independent tank type C required Control of vapour spac e within cargo tanks Vapour detection Gauging Special requirements Hydrogen 2G - - F C S ee table 2 and, either corresponding table 4 or table 5, as appropriate for the type of cargo containment systems --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCen tre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf Hydrogen 2G - - F C See table 2 and, either corresponding table 4 or table 5, as appropriate for the type of cargo con tainment systems Table 2: Special requirements for carriage of liquefied hydrogen in bul k No.

Special requirement Related hazard A-1 Requirements for materials whose design temperatu re is lower than -165°C should be agreed with the Administration, paying attention to ap propriate standards.

Where minimum design temperature is lower than -196°C, property testing for insulation m aterials should be carried out with the appropriate medium, over a range of temperatures expected in service.

Low temperature (see 4.2.1) A-2 Materials of construction and ancillary equipment such a s insulation should be resistant to the effects of high oxygen concentrations caused by condensation and enrichment at the low temperatures attained in parts of the cargo syste m (refer to the requirement for nitrogen).

This special requirement is applied --- URL: https://wwwcdn.imo.org/localresources/en/Kn owledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf low temperatures att ained in parts of the cargo system (refer to the requirement for nitrogen).

This special requirement is applied to all locations where contact with condensed oxygen is anticipated under normal conditions and foreseeable single failure scenarios.

Low temperature (see 4.2.2) A-3 For cargo pipes containing liquid hydrogen and cold hydr ogen vapour, measures should be taken to prevent the exposed surfaces from reaching -183 °C.

For places where preventive measures against low temperature are not sufficiently effect ive, such as cargo m anifolds, other appropriate measures such as ventilation which avoi ds the formation of highly enriched oxygen and the installation of trays recovering liqu id air may be permitted in lieu of the preventive measures.

Insulation on liquid hydrogen piping syst ems exposing to air should be of non -combusti ble material and should be designed to have a seal in the outer covering to prevent the condensation of air and subsequent --- URL: https://wwwcdn.imo.org/localresources/en/Kno wledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf material and should b e designed to have a seal in the outer covering to prevent the condensation of air and s ubsequent oxygen enrichment within the insulation.

Low temperature (see 4.2.2) A-4 Appropriate means, e.g.

filtering, should be provided in cargo piping systems to remove impure substances conden sed at low temperature.

Low temperature (see 4.2.3) RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INT ERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 2 0, page 5 I:\MSC\108\MSC 108-20-Add.1.docx No.

Special requirement Related hazard A-5 Pressure relief systems should be suitably design ed and constructed to prevent blockage due to formation of water or ice.

Low temperature (see 4.2.4) A-6 At places where contact with hydrogen is anticipated, su itable materials should be used to prevent any structural deterioration owing to hydroge n embrittlement and degradation of strength and --- URL: https://wwwcdn.imo.org/localres ources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf material s should be used to prevent any structural deterioration owing to hydrogen embrittlement and degradation of strength and fatigue properties due to continual exposure to hydrogen , as necessary.

Hydrogen embrittlement (see 4.3) A-7 Double tube structures ensuring no leakage, or fixe d hydrogen detectors being capable of detecting a hydrogen leak, should be provided for confined places where leakage of hydrogen may occur, such as cargo valves, flanges and s eals.

Susceptibility to leakage (see 4.4.2) A-8 Helium or a mixture of 5% hydrogen and 95% nit rogen should be used as the tightness test medium for cargo tank and cargo piping.

Susceptibility to leakage (see 4.4.3) A-9 The amount of carbon dioxide carried for a car bon dioxide fire-extinguishing system should be sufficient to provide a quantity of free gas equal to 75% or more of the gross volume of the cargo compressor and pump rooms in a ll cases.

Fire by Hydrogen (see 4.7.3) Wide range of --- URL: https://wwwcdn.imo.org/localresource s/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf the cargo com pressor and pump rooms in all cases.

Fire by Hydrogen (see 4.7.3) Wide range of flammability limits (see 4.10) A-10 When dete rioration of insulation capability by single damage is possible, appropriate safety meas ures should be adopted taking into account the deterioration.

High pressure (see 4.8) A-11 Appropriate measures should be provided to prevent vents be coming blocked by accumulations of ice formed from moisture in the air.

Low temperature (see 4.2.2) A-12 Due consideration should be given to means for handling boil-off gas.

High pressure (see 4.8) A-13 Due consideration should be given to static electricity ass ociated with rotating or reciprocating machinery including the installation of conductiv e machinery belts and precautionary measures incorporated in operating and maintenance p rocedures , in addition to the bonding of tanks, piping and equipment required by paragr aph 5.7.4 of the Code.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.566(109).pdf addition to the bonding of tanks, piping and equipment r equired by paragraph 5.7.4 of the Code.

Anti-static clothing and footwear, and a portable hydrogen detector should be provided f or each crew member working in the cargo area.

Static electricity (see 4.9.2) A-14 The cargo operation manuals required in paragraph 18 .2 of the Code should include limitations of various operations in relation to environme ntal conditions.

Wide range of flammability limits (see 4.10) RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108 /20/Add.1 Annex 20, page 6 I:\MSC\108\MSC 108-20-Add.1.docx No.

Special requirement Related hazard A-15 An appropriate procedure should be established f or warm -up, inert gas purge, gas -free, hydrogen purge and pre -cooling.

The procedure should include: .1 selection of inert gas in relation to temperature limit ; .2 measurement of gas concentration; --- URL: https://wwwcdn.imo.org/localresources/en /KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf The procedure sho uld include: .1 selection of inert gas in relation to temperature limit; .2 measurement of gas concentration; .3 measurement of temperature; .4 rates of supply of gases; .5 con ditions for commencement, suspension, resuming and termination of each operation; .6 tre atment of return gases; and .7 discharge of gases.

Prevention of dangerous purging operation (see 4.11) A-16 Only almost pure para-hydrogen (i.e.

more than 95%) should be loaded in order to avoid excessive heating by ortho - to para-h ydrogen conversion.

General (see 4.1) A-17 Fire detectors for detecting hydrogen fire should be selected, ta king into account the features of hydrogen fire, to the satisfaction of the Administrati on.

Features of hydrogen fire and fire hazard (see 4.7.4) A-18 At the design stage, dispersi on of hydrogen from vent outlets should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces, --- URL: https://wwwcdn.imo. org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109). pdf should be analysed in order to minimize risk of ingress of flammable gas into accomm odation spaces, service spaces, machinery spaces and control stations.

Extension of hazardous areas should be considered based on the results of the analysis.

Low density and high diffusivity (see 4.5) A-19 Due consideration should be given to app ropriate safety measures to prevent formation of explosive mixture in the case of a leak age and permeation of hydrogen, including: .1 installation of hydrogen detectors in orde

r to detect a possible ground -level travel of low temperature hydrogen gas, and at high points in spaces where warm hydrogen gas can be trapped; and .2 application of "best pra ctice" for land -based liquid hydrogen storage taking into account appropriate guidance such as "Cryogenics Safety Manual – Fourth Edition (1998)"8).

General (see 4.1) A-20 In the case that fusible elements are used as a means of fire --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCR esolutions/MSC.566(109).pdf Edition (1998)"8).

General (see 4.1) A-20 In the case that fusible elements are used as a means of fire det ection required by paragraph 18.10.3.2 of the Code, flame detectors suitable for hydroge n flames should be provided in addition at the same locations.

Appropriate means should be adopted to prevent the activation of ESD system owing to fal se alarm of flame detectors, e.g.

avoiding activation of ESD system by single sensor (voting method).

Fire hazard (see 4.7.4) RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, pa ge 7 I:\MSC\108\MSC 108-20-Add.1.docx No.

Special requirement Related hazard A-21 Consideration should be given to enhance the ven tilation capacity of the enclosed spaces subject to liquefied hydrogen leakage and perme ation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Ind exofIMOResolutions/MSCResolutions/MSC.566(109).pdf leakage and permeation, taking into a ccount the latent heat of vaporization, specific heat and the volume of hydrogen gas in relation to temperature and heat capacity of adjacent spaces.

Low density and high diffusivity (see 4.5) A-22 Liquid and gas hydrogen pipes should not pass through enclosed spaces in addition to other than those referred to in paragraph 5. 2.2.1.2 of the Code, unless: .1.1 the spaces are equipped with gas detection systems whi ch activate the alarm at not more than 20% LFL and shut down the isolation valves, as ap propriate, at not more than 40% LFL (see sections 16.4.2 and 16.4.8 of the Code); and .1 .2 the spaces are adequately ventilated; or .2 the spaces are maintained in an inert con dition.

Susceptibility to leakage (see 4.4) A-23 A risk assessment should be conducted to ensure that risks arising from liquefied hydrogen cargo affecting persons on board, the environ ment, the structural strength or the integrity of the ship --- URL: https://wwwcdn.imo.o rg/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).p df arising from liquefied hydrogen cargo affecting persons on board, the environment, th e structural strength or the integrity of the ship are addressed.

Consideration should be given to the hazards associated with properties of liquefied hyd rogen and hydrogen gas, physical layout, operation and maintenance, following any reason ably foreseeable failure.

For the risk assessment, appropriate methods, e.g.

HAZID, HAZOP, FMEA/FMECA, what-if analysis, etc., should be adopted taking into account IEC/ISO 31010:2019 "Risk management – Risk assessment techniques"7) and SAE ARP 5580 -20 01 "Recommended failure modes and effects analysis (FMEA) practices for non-automobile a pplications"9).

General (see 4.1) A-24 Relief valve sizing should be undertaken for the most onerous sce nario.

The evaluation should include the fire scenario and should consider the resulting magnit ude of the heat flux on the cargo containment system.

High pressure hazard (see 4.8) --- URL: https://wwwcdn.imo.org/localresources/en/Knowled geCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf should consider the resul ting magnitude of the heat flux on the cargo containment system.

High pressure hazard (see 4.8) A-25 A filling limit exceeding 98% at reference temperatu re should not be permitted.

High pressure hazard (see 4.8) A-26 Bolted flange connections of hydrogen piping should be avoided where welded connections are feasible.

Susceptibility to leakage (see 4.4.2) A-27 Due consideration should be given to the invi sible nature of hydrogen fire from the viewpoint of safety of ships and especially perso nnel in case of fire.

Fire hazard (see 4.7.1) 3 EXPLANATION ON GENERAL REQUIREMENTS 3.1 Properties of liquefie d hydrogen The application of general requirements in the Code for liquefied hydrogen ha s been considered based on a comparison study on the physical properties of liquefied hy drogen and LNG.

LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate --- URL: http s://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolution s/MSC.566(109).pdf and LNG.

LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate flammable hig h pressure gas.

For reference, table 3 shows the comparison of physical properties of hydrogen and metha ne, the major component of LNG.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 8 I:\MSC\108\MSC 108 -20-Add.1.docx Table 3: Comparison of physical properties of Hydrogen and Methane Hydrog en Methane References Boiling temperature (K)* 20.3 111.6 ISO1), Annex A, Table A.3 Liqu id density (kg/m3)* 70.8 422.5 ISO1), Annex A, Table A.3 Gas density (kg/m3)** (Air: 1.1 98) 0.084 0.668 NIST RefProp10) Viscosity (g/cm•s x 10-6) Gas Liquid 8.8 13.49 10.91 116 .79 NIST RefProp10) NIST RefProp10) Flame temperature in air (°C) 2396 2230 Calculated u sing Cantera and GRI 3.0 mechanism Maximum burning velocity (m/s) 3.15 0.385 Calculated using Cantera and GRI 3.0 mechanism --- URL: https://wwwcdn.imo.org/localresources/en/Kn owledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf Maximum burning velo city (m/s) 3.15 0.385 Calculated using Cantera and GRI 3.0 mechanism Heat of vaporizatio n (J/g)* 454.6 510.4 ISO1), Annex A, Table A.3 Lower flammability limit (% vol.

fraction)*** 4 5.3 ISO1), Annex B, Table B.2 Upper flammability limit (% vol.

fraction)*** 77 17.0 ISO1), Annex B, Table B.2 Minimum ignition energy (mJ)*** 0.017 0.2 74 ISO1), Annex B, Table B.2 Auto-ignition temperature*** 858 810 ISO1), Annex B, Table B.2 Toxicity Non Non Orange book5) Temperature at critical point (K) 33.19**** 190.55 Hy drogen: ISO1), Annex A, Table A.1 Methane: The Japan Society of Mechanical Engineers, Da ta Book, Thermophysical Properties of Fluids (1983) Pressure at critical point (kPaA) 13 15**** 4595 Hydrogen: ISO1), Annex A, Table A.1 Methane: The Japan Society of Mechanical Engineers, Data Book, Thermophysical Properties of Fluids (1983) Remarks: * At their nor mal boiling points for comparison purpose.

** At normal temperature and pressure.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.566(109).pdf Properties of Fluids (1983) Remarks: * At their normal b oiling points for comparison purpose.

** At normal temperature and pressure.

*** Ignition and combustion properties for air mixtures at 25°C and 101.3 kPaA.

**** Normal Hydrogen.

3.2 Explanation on respective requirements 3.2.1 Ship type (column 'c') As the hazard as sociated with hydrogen cargo is flammability but not toxicity, the ship type is consider ed 2G.

3.2.2 Independent tank type C required (column 'd') Independent tank type C is allocated only to dangerous goods of class 2.3 whose vapour density is heavier than air.

Independent tank type C is considered not to be required for liquefied hydrogen.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 9 I:\MSC\108\MSC 108 -20-Add.1.docx 3.2.3 Control of vapour space within cargo tank (column 'e') --- URL: htt ps://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutio ns/MSC.566(109).pdf MSC 108/20/Add.1 Annex 20, page 9 I:\MSC\108\MSC 108-20-Add.1.docx 3 .2.3 Control of vapour space within cargo tank (column 'e') Special environment controls such as drying and inerting are generally required for liquid chemical products in consi deration of the reactivity of cargo vapour and air.

As is the case for LNG, it is considered not to be necessary to apply such requirements for liquefied hydrogen.

3.2.4 Vapour detection (column 'f') Because hydrogen is flammable and non -toxic, it is appropriate to require Flammable (F) as vapour detection for liquefied hydrogen.

3.2.5 Gauging (column 'g') On the grounds that Closed (C) gauging is required, in princi

ple, for flammable or toxic cargoes, such as methane, it is considered to be appropriate to require Closed (C) gauging for hydrogen, taking into account that hydrogen has high i gnitability and a wide flammable range in air and that closed gauging is effective to pr event leakage of gases into air.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.566(109).pdf that closed gauging is effective to prevent leakage of g ases into air.

4 SPECIAL REQUIREMENTS AGAINST HAZARDS OF LIQUEFIED HYDROGEN 4.1 Hazards of liquefied hy drogen to be considered 4.1.1 The hazards related to liquefied hydrogen are low ignition energy , a wide range of flammability limits, low visibility of flames in case of fire, high flame velocity which may lead to the detonation with shockwave, low temperature and liquefaction/solidification of inert gas and constituents of air which may result in an oxygen-enriched atmosphere, high permeation, low viscosity, and hydrogen embrittlement i ncluding weld metals.

Where vacuum insulation is adopted, due consideration should be given to the possibility of untimely deterioration of insulation properties at the envisaged carriage temperature s of liquid hydrogen.

The vacuum insulation evaluation should be specified for the normal range or upper limit of cold vacuum --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Indexof IMOResolutions/MSCResolutions/MSC.566(109).pdf insulation evaluation should be specified for the normal range or upper limit of cold vacuum pressure (CVP), and loss of vacuum sh ould be defined with respect to this value.

Accordingly, effect of vacuum pressure should be taken into account at the time of desig n and testing of piping with vacuum insulation.

Supporting structure and adjacent hull structure should be designed taking into account the cooling owing to loss of vacuum insulation.

4.1.2 Hydrogen is essentially a mixture of ortho - and para-hydrogen, with an equilibriu m concentration of 75% ortho-hydrogen and 25% para-hydrogen at ambient temperature.

When liquefied at 20K, there is a slow but continuous transformation of ortho -hydrogen to para-hydrogen.

The exothermic conversion of the nuclear spin isomers of hydrogen (ortho- to para-hydrog en) may take place and the effect of the conversion may have an impact on the cooling ca pacity and relief valve capacity of the vessel's equipment.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.566(109).pdf the cooling capacity and relief valve capacity of the ve ssel's equipment.

4.1.3 Trace amounts of air will condense or solidify in an environment with liquid hydro gen possibly resulting in an unstable and explosive mixture.

Precautions should be taken to assure that the possibility of condensed air is accounted within properly secured hazard areas.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 10 I:\MSC\108\MSC 10 8-20-Add.1.docx 4.2 Low temperature hazard 4.2.1 Selection of appropriate material 4.2.1 .1 Tables 6.3 and 6.4 in the Code prescribe material selection for piping or cargo tanks whose design temperature is -165°C or higher.

According to Note 2 of table 6.3 and Note 3 of table 6.4 of the Code, the requirements f or materials whose design temperatures are lower than -165°C should be specially agreed with the Administration.

In this regard, the publication --- URL: https://wwwcdn.imo.org/localresources/en/Knowle dgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf than -165°C should be sp ecially agreed with the Administration.

In this regard, the publication by AIAA2) introduces some appropriate materials correspo nding to the design temperature and the Administration should take into account such ref erences for the material selection.

4.2.1.2 Although paragraph 4.19.3 in the Code requires testing of materials used for the rmal insulation for various properties adequate for the intended service temperature, th e minimum test temperature is -196°C.

The requirements in the Code do not refer to the normal boiling point of hydrogen, being

-253°C.

In case of carriage of liquefied hydrogen, special requirements should be provided to co nsider the lower design temperature.

4.2.2 Measures for condensed air 4.2.2.1 In the case of nitrogen whose normal boiling po int is -196°C, for which air condensation and oxygen enrichment are concerns, the follow ing special requirement has already been included in paragraph 17.17 in the Code: --- UR L: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCRes olutions/MSC.566(109).pdf condensation and oxygen enrichment are concerns, the following special requirement has already been included in paragraph 17.17 in the Code: "Material of construction and ancillary equipment such as insulation shall be resistant to the eff ect of high oxygen concentrations caused by condensation and enrichment at the low tempe ratures attained in parts of the cargo system.

Due consideration shall be given to ventilation in such areas where condensation might o ccur to avoid the stratification of oxygen-enriched atmosphere." A similar special requi rement is applicable to hydrogen.

4.2.2.2 A vent may be blocked by accumulation of ice formed from moisture in the air, wh ich may result in excessive pressure leading to rupture of the vent and relevant piping (see paragraph 4.2.4).

4.2.3 Removal of impure substances condensed The removal of impure substances, such as t hose contained in condensate in pipes, should --- URL: https://wwwcdn.imo.org/localresou rces/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf 4.2.3 Remo val of impure substances condensed The removal of impure substances, such as those conta ined in condensate in pipes, should be separately considered.

Installation of filters can be an appropriate measure and should be stipulated as a spec ial requirement.

4.2.4 Prevention of blockage due to formation of water or ice Pressure relief systems ma y become blocked due to formation of water or ice, depending on the temperature and humi dity of air, resulting from the low temperature of the cargo and its vapour (see paragra ph 4.2.2) .

Appropriate means should be provided to prevent such phenomena.

4.3 Hydrogen embrittlement 4.3.1 Selection of appropriate materials should be required t o prevent failures owing to hydrogen embrittlement.

The publication by AIAA 2) introduces some appropriate materials resistant to hydrogen e mbrittlement, and concludes that aluminium is the material least affected.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) --- URL: https://wwwcdn.imo.org/localre sources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf resista nt to hydrogen embrittlement, and concludes that aluminium is the material least affecte d.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 11 I:\MSC\108\MSC 10 8-20-Add.1.docx 4.3.2 International or national standards should be followed for the sel ection of materials for the design of liquefied and gaseous hydrogen installations in a marine environment.

4.4 Susceptibility to leakage 4.4.1 Prevention of leakage from pipes To mitigate undetec ted accumulation of hydrogen in a confined space, effective measures should be employed to reduce the possibility of leakage of hydrogen, taking its leakage characteristics int o account.

Effective measures can be double tube structures, or fixed hydrogen leak detectors in ar eas assessed as being highly hazardous with regard to hydrogen --- URL: https://wwwcdn.i mo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(10 9).pdf hydrogen leak detectors in areas assessed as being highly hazardous with regard t o hydrogen leakage.

Hydrogen leakage through welds, joints and seals is an important consideration for the d esign of hydrogen systems and an important operational issue.

4.4.2 Implementation of effective tightness test 4.4.2.1 Tightness tests for cargo tanks and cargo pipes/valves are required by paragraphs 4.20.3.2, 5.13.1 and 5.13.2.3 in the C ode respectively.

Helium or a mixture of 5% hydrogen and 95% nitrogen should be used as the medium for tig htness tests, instead of air, because hydrogen is highly susceptible to leakage.

4.4.2.2 For a hydrogen installation, the pipework should be pressure -tested at its desi gn pressure.

Consideration should be given to using oxygen-free nitrogen with a small molecule tracer gas, such as helium as the test medium and an electronic leak detec tor for identifying leaks.

4.4.3 Confirmation of appropriate operating procedure --- URL: https://wwwcdn.imo.org/lo calresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf le aks.

4.4.3 Confirmation of appropriate operating procedure Instructions/manuals containing th e operating procedures for the prevention of leakage during transport, methods for early detection in case of leakage, and appropriate measures after such events, should be prov ided.

For this, paragraph 18.3 of the Code requires that the information shall be on board and available to all concerned, giving the necessary data for the safe carriage of cargo.

In detail, the Code requires such information on action to be taken in the event of spil ls or leak, countermeasures agai nst accidental personal contact, procedures for cargo t ransfer, and emergency procedures to be on board.

With regard to the manuals on procedures for liquefied hydrogen during carriage and tran sfer operations, the requirements in the Code are applicable and no special requirement is necessary.

4.5 Low density and high diffusivity --- URL: https://wwwcdn.imo.org/localresources/en/K nowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf are applicable and no special requirement is necessary.

4.5 Low density and high diffusivity Though low density and high diffusivity of hydrogen may reduce the possibility of formation of a flammable atmosphere in open spaces, adequa te ventilation is necessary for enclosed spaces in cargo areas where formation of hydrog en -oxygen/air mixture may occur.

Paragraph 12.2 of the Code requires fixed ventilation systems or portable mechanical ven tilation for such enclosed spaces.

These requirements in the Code are applicable to liquefied hydrogen carriers and no spec ial requirement is necessary in this regard.

4.6 Ignitability 4.6.1 The Code requires electrical bonds of the piping and the cargo ta nks in paragraph 5.7.4, exclusion of all sources of ignition in paragraph 11.1.2, electr ical installations to minimize the risk of fire and explosion from flammable products in paragraph 10.2.1 and so on, in order to prevent ignition of flammable cargoes.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.566(109).pdf on, in order to prevent ignition of flammable cargoes.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 12 I:\MSC\108\MSC 10 8-20-Add.1.docx 4.6.2 The Code requires compliance with the relevant standards issued by the International Electrotechnical Commission (IEC) and the IEC standards specify the de tails of such safety measures depending on the respective properties of flammable gases including hydrogen.

No special requirement is necessary with regard to ignitability of hydrogen.* 4.7 Fire h azard 4.7.1 Safety of personnel in case of fire To avoid the effects of flame and UV rad iation produced by a hydrogen fire, it is effective to use fire-fighter's outfits and pr otective equipment.

The Code already requires fire -fighter's outfits for ships carrying flammable products in paragraph 11.6.1 and safety equipment in --- URL: https://wwwcdn.imo.org/localresourc es/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf outfits for ships carrying flammable products in paragraph 11.6.1 and safety equipment in paragraph 14.3.

This issue should be considered as the matter of cargo information required by paragraph 18.3 of the Code.

Due consideration should be given to the invisible nature of hydrogen fire.

4.7.2 Compatibility of fire-extinguishing systems Dry chemical powder fire -extinguishin g or carbon dioxide fire -extinguishing systems are considered to be effective in case o f hydrogen fire and such fire -extinguishing systems are already required by paragraphs

11.4 and 11.5 of the Code.

Special requiremen ts for installation of other types of fire -extinguishing systems are considered unnecessary, except with regard to the increased of amount of carbon dioxide required, as mentioned in the next paragraph in this document.

4.7.3 Increase of the amount of gas for carbon dioxide fire-extinguishing systems 4.7.3. 1 Paragraph 11.5.1 of the Code requires as follows: --- URL: https://wwwcdn.imo.org/loca lresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf 4.7. 3 Increase of the amount of gas for carbon dioxide fire-extinguishing systems 4.7.3.1 Pa ragraph 11.5.1 of the Code requires as follows: "Enclosed spaces meeting the criteria of cargo machinery spaces in 1.2.10, and the cargo motor room within the cargo area of any ship, shall be provided with a fixed fire-extinguishing system complying with the provis ions of the FSS Code and taking into account the necessary concentrations/application ra te required for extinguishing gas fires." 4.7.3.2 Chapter 5 of the FSS Code, i.e.

Fixed gas fire -extinguishing systems, requires that the quantity of carbon dioxide for cargo spaces, unless otherwise provided, shall be sufficient to give a minimum volume of free gas equal to 30% of the gross volume of the largest cargo space to be protected in the ship, in paragraph 2.2.1.1.

4.7.3.3 On the other hand, NFPA 123) requires that the design quantity of carbon dioxide for --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResoluti ons/MSCResolutions/MSC.566(109).pdf 4.7.3.3 On the other hand, NFPA 123) requires that t he design quantity of carbon dioxide for hydrogen fire should be 75% or more of the gros s volume of the protected space.

The special requirement for an increased amount of carbon dioxide should be provided for carbon dioxide fire-extinguishing systems.

4.7.4 Features of hydrogen fire Hydrogen burns at high temperature, but generally gives off less radiant heat than propane or other hydrocarbons (e.g.

only about 10% of that radiated by an equal -sized propane flame).

Although the heat radiated by a hydrogen flame is also relatively low compared to hydroc arbons, it is important to take into account the differences in heats of combustion, * E lectrical equipment used in hydrogen/air mixture should be, at least, the type of "II-C" and "T-1" as the group based on the maximum experimental safe gap for flameproof enclosu res and the temperature class based --- URL: https://wwwcdn.imo.org/localresources/en/Kn owledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf based on the maximum experimental safe gap for flameproof enclosures and the temperature class based on maxim um surface temperature, respectively, according to ISO/IEC 80079-20-14).

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 13 I:\MSC\108\MSC 10 8-20-Add.1.docx burning rate and flame size.

Hydrogen flames are colourless or nearly colourless.

Both of these characteristics make it more difficult to detect a hydrogen fire.

Even relatively small hydrogen fires are very difficult to extinguish.

The only reliable approach to extinguish a fire is to shut off the source of hydrogen su pply.

4.8 High pressure hazard 4.8.1 High pressure is a hazard common to hydrogen and other fl ammable gases listed in the Code.

To prevent overpressure, the Code requires various measures such as pressure --- URL: ht tps://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResoluti ons/MSC.566(109).pdf the Code.

To prevent overpressure, the Code requires various measures such as pressure control and pressure design.

Specifically, paragraph 8.2, in regard to the provision of pressure control of cargo tan ks, requires fittings of pressure relief valves to the cargo tanks.

Furthermore, paragraph 7.1.1 requires temperature control by the use of mechanical refri geration and/or design to withstand possible increases of tempera ture and pressure.

In addition, paragraph 15.2 specifies the filling limit of cargo tanks taking into accou nt cargo volume increase by its thermal expansion.

These requirements are applicable for hydrogen and no special requirement is considered necessary in this regard.

4.8.2 Boil-off may be a more significant issue for hydrogen than for LNG in particular w
hen insulation properties have deteriorated.

Means of handling boil -off gas should be carefully considered taking into account the f
ollowing issues: --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Index
ofIMOResolutions/MSCResolutions/MSC.566(109).pdf insulation properties have deteriorated
.

Means of handling boil -off gas should be carefully considered taking into account the f
ollowing issues: .1 Re-liquefaction of hydrogen involves very specific and costly equipm
ent.

Cargo cooling in order to avoid boil-off shows the same kind of issues; and .2 Notwithst
anding the provision in paragraph 7.4.1 of the Code, thermal oxidation of hydrogen may b
e permitted in accordance with paragraph 1.3 of the Code.

4.8.3 The special requirements in these aspects are considered necessary.

4.9 Health hazard 4.9.1 Human safety concern under low temperature With regard to the in
fluences of cold hydrogen on persons' bodies, suitable protective equipment is effective
.

In this aspect, paragraph 14.1 of the Code requires suitable protective equipment taking
into account the character of the products, therefore, no special requirement is conside
red necessary.

4.9.2 Static electricity --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCent
re/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf equipment taking into account t
he character of the products, therefore, no special requirement is considered necessary.

4.9.2 Static electricity Hydrogen ignition energy is very low and hydrogen can be easily
ignitable by static electricity and due consideration should be given to this issue, in
accordance with the requirement in the Code on suitable protective equipment.

4.9.3 Oxygen depletion and asphyxiation Leakage of hydrogen may cause low level of oxyge
n and associated asphyxiation.

4.10 Wide range of flammable limits 4.10.1 Extinguishing hydrogen fire 4.10.1.1 As menti
oned in paragraph 4.6, for flammable products the Code already requires elimination of s
ources of ignition, including use of electrical installations of appropriate types in or
der to minimize the risk of fire and explosion.

No special requ irement is considered necessary with regard to ignitability of hydrogen.

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sources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf necessa
ry with regard to ignitability of hydrogen.

RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMMENDATIONS FOR CAR
RIAGE OF LIQUEFIED HYDROGEN IN BULK MSC 108/20/Add.1 Annex 20, page 14 I:\MSC\108\MSC 10
8-20-Add.1.docx 4.10.1.2 Furthermore, with regard to the wide range of flammable limits
of hydrogen, the increased quantities of carbon dioxide as a fire -extinguishing medium
should be specified as mentioned in paragraph 4.7.

No additional special requirement is considered to be necessary with regard to the wide
range of flammable limits of hydrogen.

4.10.2 Disposal of cold hydrogen gas The wide flammability range makes disposal of cold
hydrogen gas a major hazard.

Cold plumes of released hydrogen may impede adequate dilution of hydrogen down to below
4% and may lead to flash-back to the vent from distant ignition sources outside safety -
controlled areas.

The low ignition energy and wide flammable range may present significant challenges.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.566(109).pdf areas.

The low ignition energy and wide flammable range may present significant challenges.

4.11 Prevention of dangerous purging operation 4.11.1 During cargo operations for mainte
nance, pipes and tanks should be purged with an inert gas or inert gases as illustrated
in the figure below.

For safety, due consideration should be given to temperature and boiling points of the i
nert gases.

Residual pockets of hydrogen or the purge gas will remain in the enclosure if the purgin

g rate, duration, or extent of mixing is too low.

Therefore, reliable gas concentration measurements should be obtained at a number of dif ferent locations within the system for suitable purges.

Temperature should also be measured at a number of locations.

Oxidizing agents may exist in a hydrogen containing equipment, specifically: air, cold b ox atmospheres containing air diluted with nitrogen, or oxygen-enriched air that can be condensed on process pipe work within the cold box in special circumstances.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.566(109).pdf oxygen-enriched air that can be condensed on process pip e work within the cold box in special circumstances.

4.11.2 There are special measures that may need to be put in place in order to mitigate the hazards, e.g.

air should be eliminated by nitrogen purge prior to introduction of hydrogen into cargo piping or processing equipment.

Nitrogen should then be eliminated by hydrogen purge, where there is a possibility of it s solidification in the subsequent process.

Figure 1 References in part A 1 ISO/TR 15916:2015, Basic consideration for the safety of hydrogen systems (ISO) 2 American Institute of Aeronautics and Astronautics, "Safety Sta ndard for Hydrogen and Hydrogen Systems (Guide to Safety of Hydrogen and Hydrogen System s)", 2005 (AIAA) RESOLUTION MSC.565(108) (adopted on 24 May 2024) REVISED INTERIM RECOMM ENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK --- URL: https://wwwcdn.imo.org/loc alresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.562(108).pdf RES OLUTION MSC.562(108) (adopted on 23 May 2024) STANDARDS OF TRAINING, CERTIFICATION AND W ATCHKEEPING FOR FISHING VESSEL PERSONNEL CODE (STCW-F CODE) - 1 - RESOLUTION MSC.562(108) (adopted on 23 May 2024) STANDARDS OF TRAINING, CERTIFICATION AND WATCHKEEPING FOR FIS HING VESSEL PERSONNEL CODE (STCW-F CODE) THE MARITIME SAFETY COMMITTEE, RECALLING Articl e 28(b) of the Convention on the International Maritime Organization concerning the func tions of the Committee, RECOGNIZING the importance of establishing detailed mandatory st andards of competence and other mandatory provisions necessary to ensure that all fishin g vessel personnel shall be properly educated and trained, adequately experienced, skill ed and competent to perform their duti